

Shansan Gong

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The University of Hong Kong

Summary

PhD candidate at HKU, supervised by **Lingpeng Kong**. I work on **text diffusion models** for conditional generation, reasoning, and scaling, and develop their next generation abilities as diffusion language models (**DLMs**). I'm also interested in **long-context modeling** and building personalized, tool-augmented systems.

Education

PhD in Computer Science	2023.09 – Present (estimated: 2027.01)	The University of Hong Kong (HKU)	Hong Kong
M.Sc. in Computer Science (GPA 3.78)	2019.06 – 2022.03	Shanghai Jiao Tong University (SJTU)	Shanghai
B.Eng. in Information Engineering (GPA 3.84)	2015.09 – 2019.06	Shanghai Jiao Tong University (SJTU)	Shanghai

Research Experience

Research Intern — Code Generation with DLMs; RL Post-training	2025.01 – 2025.07	Apple	Seattle, US
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- **Masked diffusion models for code**: analyzed generation order vs. accuracy/diversity in DLMs for code; trained 7B-scale diffusion coder with competitive performance to AR coders; studied augmenting the mask token with continuous space.
- **RL post-training for DLMs**: introduced a coupled-sampling GRPO variant tailored to masked diffusion models.

Research Intern — Reasoning & Scaling of Diffusion LMs	2023.11 – 2024.10	Tencent AI Lab	Shenzhen
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- **CoT for diffusion LMs**: proposed/implemented diffusion-native CoT (DoT: reasoning steps diffusing over timesteps); evaluated on arithmetic/logic/math reasoning with self-consistency variants.
- **Scaling via AR → Diffusion adaptation**: developed continual-pretraining pipelines to convert AR models (GPT/LLaMA scales) into diffusion LMs; ran ablations on training tokens/inference speed vs. quality.

Researcher — Diffusion Text Generation; Long-Context Modeling	2022.05 – 2023.08	Shanghai AI Lab	Shanghai
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- **Initial exploration of text diffusion for Seq2Seq**: designed classifier-free training and decoding recipes for conditional generation; established connections among AR/NAR/diffusion paradigms in practice.
- **Acceleration & bridging discrete/continuous spaces**: integrated ODE-based samplers and a soft absorbing state to speed up training and sampling for diffusion text generation.
- **Long-context modeling**: contributed to scaling/evaluation for long-context LMs (data processing, chunking strategies, evaluation harnesses), with takeaways later reused by collaborators.

Selected Publications * equal contribution

Huangjie Zheng, **Shansan Gong**, Ruixiang Zhang, Tianrong Chen, Jiatao Gu, Mingyuan Zhou, Navdeep Jaitly, Yizhe Zhang, “Continuously augmented discrete diffusion model for categorical generative modeling”. arXiv, 2025. [\[paper\]](#)

Shansan Gong, Ruixiang Zhang, Huangjie Zheng, Jiatao Gu, Navdeep Jaitly, Lingpeng Kong, Yizhe Zhang, “DiffuCoder: Understanding and Improving Masked Diffusion Models for Code Generation”. arXiv, 2025. [\[paper\]](#) [\[code\]](#) (700+ ★)

Shansan Gong^{*}, Shivam Agarwal^{*}, Yizhe Zhang, Jiacheng Ye, Lin Zheng *et al.*, “Scaling Diffusion Language Models via Adaptation from Autoregressive Models”. ICLR, 2025. [\[paper\]](#) [\[code\]](#) (300+ ★)

Jiacheng Ye^{*}, **Shansan Gong**^{*}, Liheng Chen^{*}, Lin Zheng *et al.*, “Diffusion of Thoughts: Chain-of-Thought Reasoning in Diffusion Language Models”. NeurIPS, 2024. [\[paper\]](#) [\[code\]](#) (170+ ★)

Chenxin An, **Shansan Gong**, Ming Zhong *et al.*, “L-Eval: Instituting Standardized Evaluation for Long Context Language Models”. ACL, **Outstanding Paper**, 2024. [\[paper\]](#) [\[code\]](#) (380+ ★)

Shansan Gong, Mukai Li, Jiangtao Feng, Zhiyong Wu, Lingpeng Kong, “DiffuSeq-v2: Bridging Discrete and Continuous Text Spaces for Accelerated Seq2Seq Diffusion Models”. EMNLP Findings, 2023. [\[paper\]](#) [\[code\]](#)

Shansan Gong, Mukai Li, Jiangtao Feng, Zhiyong Wu, Lingpeng Kong, “DiffuSeq: Sequence to Sequence Text Generation With Diffusion Models”. ICLR, 2023. [\[paper\]](#) [\[code\]](#) (800+ ★)

Honors & Awards

HKU Presidential PhD Scholarship (HKU-PS)		ACL 2024 Outstanding Paper		SIGIR 2022 Student Travel Award
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Service

Conference Reviewer: ACL; EMNLP; NeurIPS; ICLR; ICML; ARR; COLM

Open Source: Primary maintainer/contributor on [DiffuSeq](#), [DiffuLLaMA](#), [DiffuCoder](#) repositories